

Pheast Formal Status Report

1. Project Title: CD24 Expression in normal pediatric tissues and tumors

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2. Introduction / Background:

What was the major goal of this consulting collaboration? Expand to biological questions, some introduction to the goals of the client

My individual project within the larger Pheast capstone group focused on exploring whether PHST001, which is currently in clinical trials for adult solid tumors, may also have future relevance in pediatric cancer contexts. Since PHST001 is connected to the CD24 immune-evasion pathway, my project used public pediatric single-cell and bulk RNA-seq datasets to examine where CD24 and related markers, such as SIGLEC10, are expressed.

The goal was to identify pediatric tissues, cell types, or tumor contexts where CD24 expression may be biologically relevant. Through this work, I generated organized code, cleaned visualizations, and interpretable results to help Pheast consider possible future directions for PHST001 beyond adult solid tumors.

3. Objectives / Research Questions:

What were the specific project goals (bullet points are fine):

- Characterize the expression patterns of CD24, SIGLEC10, in normal pediatric peripheral blood using publicly available scRNA-seq data
- Expand the characterization of CD24 and SIGLEC10 expression across normal pediatric tissues, including bone, thyroid, thymus, and brain/spinal cord, using publicly available scRNA-seq data.
- Evaluate whether CD24 expression is associated with overall survival in pediatric cancers using publicly available bulk RNA-seq and clinical survival data.

Week 1–2: Project Onboarding and Data Preparation

- ✓ Initial meetings with Grace, Josh, and Suzana.
- ✓ Reviewed the overall project focus and learned how public scRNA-seq datasets can be used to study gene expression patterns.
- ✓ Began identifying available public pediatric healthy blood datasets for Aim 1.

Week 3–5: Public Dataset Review and scRNA-seq Analysis

- ✓ Learned how to search for and evaluate public scRNA-seq datasets for pediatric healthy blood and healthy tissue analysis.
- ✓ Learned how to search for and evaluate public scRNA-seq datasets.
- ✓ For Aim 1, reviewed pediatric healthy blood datasets and performed basic filtering and QC checks.
- ✓ For Aim 2, identified healthy tissue datasets, including bone, brain, thyroid, and thymus.
- ✓ Reviewed dataset details such as dataset source, associated papers, donor information, age groups, cell counts, and annotations.
- ✓ Began preliminary visualizations, including early heatmaps and UMAPs.

Week 6–8: Final Visualizations, Aim 3 Analysis, and FRM Slide Preparation

- ✓ Created final visualizations for Aim 1, including heatmaps and UMAPs.
- ✓ Created final visualizations for Aim 2 using healthy tissue scRNA-seq datasets.
- ✓ Started and completed Aim 3 bulk RNA-seq analysis.
- ✓ Generated Aim 3 figures and summaries for cancer datasets.
- ✓ Prepared and updated the FRM slide deck with methods, dataset descriptions, results, and visualizations.
- ✓ Checked and verified code, figures, and analysis steps with Josh for the FRM presentation.

Week 9–10: Reporting and Handoff

- ☑ Finalize the FRM slide deck and prepare for the final presentation on June 5th 2026.
 - ☑ Revise slides based on feedback from Grace, Rachel, and Josh
 - ☑ Complete the final report with methods, results, interpretations, and documentation.
 - ☑ Organize final Quarto documents, code, figures, and worklog entries for submission.
 - ☑ Hold final meeting with Pheast for knowledge transfer.
 - ☑ Provide recommendations for next steps and future analysis.
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4. Data & Tools

Datasets used in this project included NCBI Gene Expression Omnibus, Chan Zuckerberg CELLXGENE discover, cBioPortal, and Human Cell Atlas. The project used scRNAseq data and Bulk-RNAseq data containing donor, clinical, and overall survival data. Most of the datasets were already normalized by a factor of logCP10K. scRNAseq dataset were mainly done by the Seurat package and using other R libraries such as dplyr, and tidyr. Bulk-RNAseq datasets were all checked to have clinical metadata and overall survival data to run Kaplan Mayer analysis.

R packages and Tools:

- Seurat, SingleCellExperiment, Matrix, GEOquery, and zellkonverter for loading and processing scRNA-seq data
- dplyr, tidyr, and ggplot2 for metadata filtering, organization, and visualization
- ComplexHeatmap and circlize for heatmap generation
- survival and survminer for Kaplan–Meier curves, log-rank tests, and Cox regression
- cBioPortal/OpenPedCan API for accessing pediatric tumor expression and survival data

SharePoint drive: [Pheast Capstone project deliverables - Kelly Nguyen](#)

5. Methods Overview

Project 1&2: Healthy Pediatric scRNA-seq Tissue Analysis

1. I used publicly available pediatric single-cell RNA sequencing datasets from Human Cell Atlas, Chan Zuckerberg CELLxGENE Discover, and GEO. The datasets represented healthy or adjacent normal pediatric tissues, including peripheral blood, thymus, growth plate/bone, brain, and thyroid.
2. For each dataset, we filtered samples to focus on healthy pediatric donors or the closest available normal pediatric tissue group. For thyroid, we used children/adolescent/young-adult adjacent normal thyroid samples from tissue region P. For growth plate, we used healthy human pediatric growth plate samples.
3. I investigated the clinical metadata for each dataset to identify donor/sample IDs, age or developmental stage, tissue source, disease/normal status, and available cell-type labels. When datasets already included cell-type annotations, those labels were used directly for heatmaps and UMAPs.
4. I performed dataset quality assessment using detected gene counts, mitochondrial percentage, ribosomal/healthy-cell profile, read depth, and minimum cell numbers per cell type. Low-quality cells were removed before downstream visualization.
5. I checked whether each dataset was already normalized. If only raw count layers were present, we normalized expression using $\log_{10}(\text{counts per } 10k)$. If normalized expression was already provided, we used the existing normalized assay to avoid double-normalizing the data.
6. I focused on candidate genes related to CD24 signaling and tumor/immune biology, mainly CD24 and SIGLEC10, with additional genes such as NECTIN4, CDH1, ST8SIA1, and B4GALNT1 included when appropriate.
7. For datasets without detailed author-provided cell-type labels, such as thyroid and growth plate, we assigned broad cell identities using known marker genes. These marker-based labels were used to identify major tissue, stromal, vascular, and immune cell populations.
8. I averaged normalized gene expression by cell type or marker-derived cell group. This allowed us to compare CD24 and SIGLEC10 expression across different pediatric tissues and cell populations.

9. I generated broad and detailed heatmaps to visualize expression patterns across cell types. Heatmaps used quantile-based color scaling and included donor and cell counts to show both expression level and dataset support for each cell group.

10. I used these analyses to identify where CD24 and SIGLEC10 were most strongly expressed across healthy pediatric tissues, including blood immune cells, thymus cell populations, thyroid epithelial/immune populations, growth plate chondrocytes and immune cells, and brain cell types.

Project 3: OpenPedCan Dataset Overall Survival Analysis

1. Used the OpenPedCan pediatric cancer dataset through the cBioPortal portal. This dataset provided RNA-seq expression data and clinical survival information for pediatric and young adult tumor samples.

2. Extracted CD24 RNA expression values from the OpenPedCan RNA-seq molecular profile and merged them with clinical data containing cancer diagnosis, patient age, overall survival time, and survival status.

3. Filtered the dataset to include tumor samples from pediatric and young adult patients with medulloblastoma, osteosarcoma, Neuroblastoma, and Ewing sarcoma. Samples were kept if they had available CD24 expression, age information, overall survival time, and survival status.

4. Converted overall survival information into survival months and event status. Patients listed as deceased were coded as events, while living patients were treated as censored observations.

5. For each cancer type, tumors were stratified into CD24 high and CD24 low groups using the median CD24 expression value within that tumor type. Samples with CD24 expression greater than or equal to the median were assigned to the CD24 high group, while samples below the median were assigned to the CD24 low group.

6. Generated Kaplan–Meier survival curves to compare overall survival between CD24 high and CD24 low tumors for each cancer type. Each plot included a risk table showing the number of patients remaining at each time point.

7. Did log-rank tests to determine whether survival differed significantly between the CD24 high and CD24 low groups within each tumor type.

8. Ran Cox proportional hazards regression models using continuous CD24 expression to test whether CD24 predicted overall survival. This allowed us to evaluate CD24 as a continuous survival-associated variable rather than only as a high/low group.

6. Figures

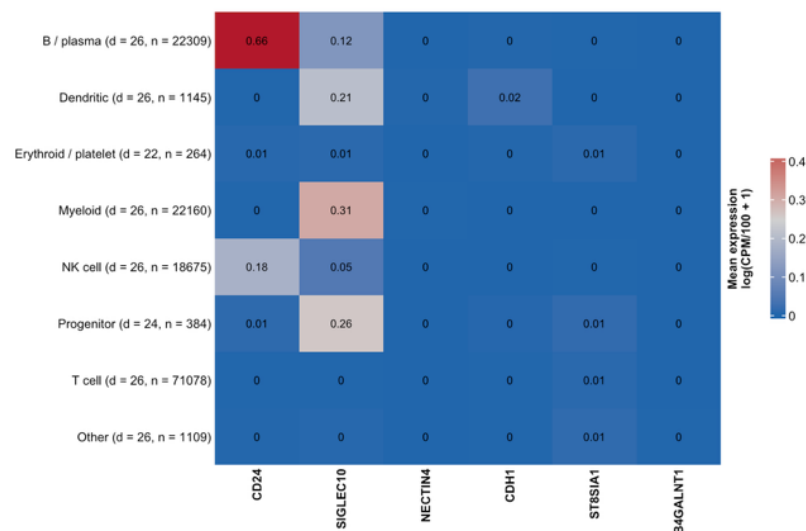


Figure1. Mean normalized expression of CD24, SIGLEC10, NECTIN4, CDH1, ST8SIA1, and B4GALNT1 across healthy pediatric peripheral blood cell populations. CD24 expression is highest in B/plasma cells, while SIGLEC10 is most enriched in myeloid and dendritic populations.

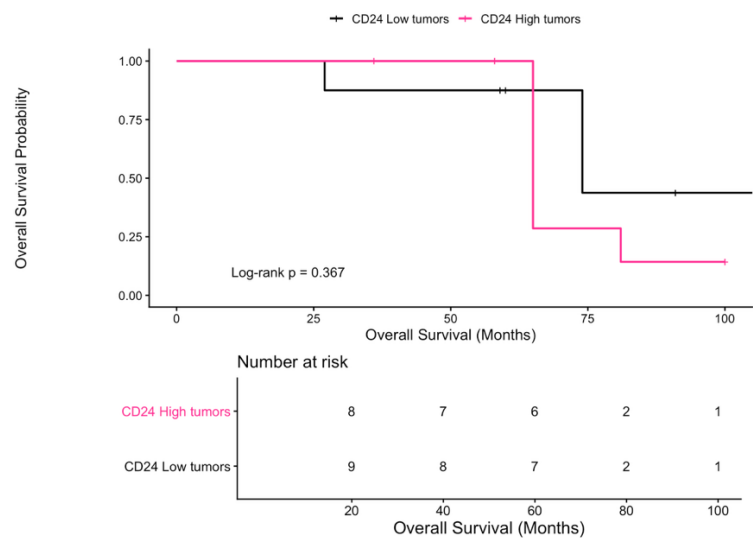


Figure 2. Kaplan Meier survival analysis comparing CD24-high and CD24-low Ewing sarcoma tumors from OpenPedCan. CD24-high tumors showed a lower survival trend, but the difference was not statistically significant by log-rank test.

7. Conclusions & Findings

This project found that overall CD24 expression is tissue and cell type dependent across healthy pediatric tissues. In peripheral blood, CD24 was most enriched in B/plasma cells, while SIGLEC10 was more enriched in myeloid and dendritic populations. NECTIN4, CDH1, ST8SIA1, and B4GALNT1 showed little to no expression across most healthy blood cell types, suggesting that these genes may be more relevant in epithelial, solid tissue, or tumors rather than being involved in immune cells.

Across the additional healthy tissue datasets, CD24 expression was observed in thyrocytes, thymic B-lineage cells, growth plate chondrocytes, and dentate gyrus neurons, while SIGLEC10 was most notable in microglia. But, for the thyroid dataset specifically, the exact donor ages were not provided, making it difficult to determine whether or not the observed expression patterns are fully representative of normal pediatric thyroid tissue. In addition, all of the collected thyroid samples within the dataset were tumor-adjacent tissues, making it unclear whether CD24 would normally be expressed at similarly high levels in healthy pediatric thyroid tissue without the influence of a nearby tumor. These findings suggest that CD24 biology may differ depending on tissue context, although the influence of developmental stage cannot be fully assessed from the available thyroid data.

The OpenPedCan survival analyses showed that high CD24 expression was significantly associated with worse overall survival across all pediatric tumors and was strongest in neuroblastoma when we looked into 3 specific tumors from previous analysis findings done by Josh. However, CD24 was not significantly associated with survival in medulloblastoma, osteosarcoma, or Ewing sarcoma, suggesting that CD24 may have tumor-type-specific relevance rather than acting as a universal survival marker.

Moving forward, these findings support expanding CD24 expression analysis into additional normal pediatric tissues that are important for development, such as reproductive tissues. Future work should also examine more into specific medulloblastoma subgroups more closely to understand why CD24 was not significantly associated with survival in that dataset. Additional normal pediatric thyroid datasets would help determine whether the high CD24 expression observed in tumor-adjacent thyroid tissue reflects true healthy pediatric thyroid expression.